

SPRUCE PLANTATION EFFECTS ON SOIL MOISTURE AND CHEMICAL ELEMENT DISTRIBUTION

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(Accepted 25 April 1980)

ABSTRACT

Messenger, A.S., 1980. Spruce plantation effects on soil moisture and chemical element distribution. *Forest Ecol. Manage.*, 3: 113–125.

Conflicting opinions are extant as to whether spruce has a favorable or unfavorable effect on soil fertility. This controversy was addressed at the Morton Arboretum in north-eastern Illinois by collecting and comparing data on several dynamic and semi-static soil properties in a 48-year-old mixed spruce plantation with those from the surrounding native hardwood forests on the same soil type. Throughout May, the surface soil was drier in the spruce plantation. The entire B horizon, however, was significantly more moist in the spruce plantation than in the hardwood forests from late spring to fall. In the spruce plantation only, the dense, calcareous C horizon began to dry out rapidly in mid-June to a depth of 1.5 m and continued unabated until late summer despite surface soil moisture recharge episodes.

The surface soil in the spruce plantation was significantly more acid and exhibited a significantly higher concentration of extractable manganese than the hardwood forest surface soil. Differential biocycling intensities are suggested as partial explanations of these chemical differences.

Soils of the type studied are well suited to *Picea abies* compared to other species commonly planted on them. One rotation of *Picea abies* intermixed with *Picea pungens* may render the soil more suitable for other species which would be hindered by a less acid surface soil and a dense, calcareous C horizon not containing numerous old root channels formed by the previous generation of trees.

INTRODUCTION

Numerous studies in Europe and Russia have assessed the influence of spruce plantations on soil nutrient availability and soil moisture regimes. Despite this attention, conflicting opinions still exist as to whether soil conditions improve or deteriorate as influenced by the species involved. Part of the controversy stems from inadequate perception of inherent soil heterogeneity and other conditions antecedent to plantation establishment (Stone, 1975).

This paper compares data on several dynamic and semi-static soil proper-

ties in a spruce plantation on Morley silt loam (fine, illitic, mesic family of Typic Hapludalfs) with those in the surrounding native hardwood forest on the same soil type. It suggests reasons for the observed differences and discusses silvicultural and arboricultural implications.

LITERATURE REVIEW

Evergreen conifers transpire over a longer season than deciduous trees, and precipitation interception losses continue throughout the year. Thus, available soil moisture is often reduced earlier under evergreen forests and may average less throughout the growing season, requiring a longer period of recharge (Douglass, 1967). Such a relationship makes it difficult to accept the numerous contentions that perched water-tables are raised and/or prolonged by evergreen tree plantation establishment. One investigator found evidence of greater dryness in the surface horizons of some soils planted to Norway spruce (*Picea abies*) compared to comparable horizons and soils within a beech forest; lower horizons, however, were more moist (Mitscherlich, 1970).

Rooting patterns drastically affect soil moisture regimes. Norway spruce is generally considered to have shallow roots, but several studies contradict this generalization. Evidence of moisture uptake by this species to depths of greater than 1 m have been reported (Blume et al., 1969). The explanation for this apparent contradiction probably lies in the morphological plasticity of the Norway spruce root system. On sandy soils it roots deeply as Scots pine; on poorly drained soils it roots shallowly; and in some soils it exhibits a bi-morphic or two-layered pattern, one layer near the surface joined by un-branched sinker roots to a lower layer which may extend to a depth of 1.5 m (Köstler in Mitscherlich, 1970).

Most research indicates high acidity and low base saturation of humus layers under spruce plantations, influenced, at least in part, by the spruce litter (Stone, 1975). The enhancement of iron oxide removal from the soil has been attributed to the release of chelating agents from Norway spruce trees (Epstein, 1972). The effect of spruce on trace elements and macronutrients other than those controlling base saturation (i.e., Ca, Mg, and K) is not clear. For example, many studies have emphasized the effect of spruce plantations on available soil nitrogen. Some of these point to excessive immobilization of nitrogen during spruce litter decomposition, while others suggest that spruce has the capacity to increase supplies of soil nitrogen available for uptake (Fisher and Stone, 1969).

MATERIALS AND METHODS

In 1971, a pilot soil moisture monitoring program was initiated to identify possible plant community effects on the soil moisture regime. Core samples were taken for gravimetric moisture determination, and nuclear probe access tubes were installed in three hardwood forest communities and one spruce

plantation in the Morton Arboretum (37 km west of Chicago) on Morley silt loam or soils of similar texture and drainage developed from essentially identical glacial till.

Three soil characterization pits 1.5 m deep were excavated in each forest community. The exposed soil profiles were described and photographed. Each profile within a community differed somewhat from others in its precise horizon composition and in the depths and thicknesses of common horizons. Because of this complexity, generalized patterns of horizons and root distributions within them are presented as composites of the spruce site profiles and the hardwood site profiles (Table I).

Samples for physical and chemical analyses were taken from each horizon of each profile. Soil reaction was determined on field moist samples; exchangeable cations and pH 4.8 ammonium acetate-extractable Mn, Fe, and Al were determined on air-dried samples (McLean et al., 1960; Cox and Kamprath, 1972). Duplicate samples at three depths within the upper 40 cm of soil at each site were taken bi-weekly from 15 June through 14 October 1972. Each sample was sub-divided for inorganic nitrogen determinations, and oven-dry weight and moisture calculations. On each sampling day, $\text{NH}_4\text{-N}$ was extracted with Morgan's solution and determined by nesslerization; $\text{NO}_3\text{-N}$ was extracted with distilled water and determined by the nitrate reductase method (Peech and English, 1944; Lowe and Hamilton, 1967).

Hardwood leaves were collected near the soil pits in early September 1972, from two sugar maples (*Acer saccharum*), three northern red oaks (*Quercus rubra*), and two white oaks (*Quercus alba*), more emphasis being placed on red oak because of greater proportional abundance. Spruce needles were collected in October 1972, from five Colorado blue spruces (*Picea pungens*) and five Norway spruces (*Picea abies*). Only the old needles were dried at 70°C, ground, and analyzed by emission spectrograph and micro-Kjeldahl (for N) methods, since the composition of these needles was considered to have more pedological significance than that of the needles originated in 1972.

Two of the native hardwood communities reported on here can be classified as "northern hardwood" since the overstory is dominated by sugar maple and northern red oak with lesser representation by linden (*Tilia americana*), black walnut (*Juglans nigra*), and white ash (*Fraxinus americana*). The third native hardwood community is best classified as white oak—red oak; white oak accounts for more than 50% of the overstory trees throughout the community. Overstory species other than these two are negligible and widely scattered. The spruce community is a plantation of Norway spruce and Colorado blue spruce established in the fall of 1923 on a field which had been devoted to a rotation of corn, wheat, soybeans, and alfalfa for at least the previous 22 years. The earliest possible cropping was in the mid-eighteenforties. Little evidence of erosion was noted at the time of plantation establishment. Understory vegetation at the white oak—red oak study site is abundant and typical of the community, consisting of small trees, shrubs, vines, and forbs. In contrast, the understory at the northern hardwood site consists largely of young trees

TABLE I

Horizonation and root distribution in soil profiles at spruce and hardwood sites

Spruce site profiles			Hardwood site profiles		
Horizons	Ave. depth range (cm)	Roots	Horizons	Ave. depth range (cm)	Roots
Ap	0-15	All sizes up to 50 mm abundant	A1, A11	0-9	Very abundant up to 15 mm diam., few up to 40 mm
B1, B21	15-35	Abundant; all < 15 mm; mostly < 10 mm	A12, A2, A3	9-17	Abundant up to 15 mm diam., few up to 25 mm diam.
B2, B3	35-76	Few; mostly < 5 mm	B1, B&A, B21	17-27	Numerous up to 15 mm diam., few up to 25 mm diam.
C	76-150	Few; all < 5 mm and extending downward to at least 1.5 m within till fractures only	B1, B2, B22	27-58	Numerous up to 15 mm diam.; few up to 25 mm diam.; except many up to 50 mm diam. at white oak site
			B3	58-86	Numerous up to 15 mm diam.; almost none larger except up to 25 mm diam. at white oak site
			C	86-150	Few to numerous up to 15 mm diam.; none larger; confined to till fractures at white oak site

(mostly sugar maple) and forbs. The spruce site is essentially devoid of under-story.

RESULTS

Moisture measurements made in 1971 yielded the data shown in Fig. 1 which point to four major differences in soil moisture regimes between the spruce plantation and the hardwood forests: (1) soil moisture was distinctly and consistently less in the upper 10 cm of soil at the spruce site until mid-June; (2) fluctuations in the upper 10 cm of soil were much more marked at the hardwood sites after mid-June; (3) the soil moisture content at the 15–75 cm depth was distinctly and consistently greater at the spruce site; and (4) moisture depletion at the 120–150 cm depth was more rapid and much greater at the spruce site from mid-June through mid-August.

The lower mean values of foliar N and Mg in spruce foliage were significantly different from those of the hardwoods and the high foliar Mn of Norway spruce was significantly different from those of both Colorado blue spruce and the hardwoods (Table II).

The soil horizons at the hardwood sites coinciding in depth with the Ap horizon of the spruce site were higher in pH, $\text{NO}_3\text{-N}$, and exchangeable Ca, but lower in extractable Al, Mn, $\text{NH}_4\text{-N}$, Mg, and exchangeable H than at the spruce site. Among these differences, however, statistical significance can only be assigned to pH and extractable Mn (Table III). Horizons at the 25 cm depth showed no significant differences in any of the above constituents. When the entire 10–100 cm soil zones are compared, only extractable manganese is significantly different, the higher values at the spruce site showing secondary maximum differences at 40–75 cm (Fig. 2).

TABLE II

Comparative concentrations of chemical elements in spruce and hardwood foliage

	N (%)	K (%)	Ca (%)	Mg (%)	Mn (ppm)	Al (ppm)
Hardwoods <i>n</i> =7 (2 maple, 3 red oak, 2 white oak)	2.02	0.68	1.51	0.30	397	99
Colorado blue spruce <i>n</i> =5	1.05*	0.69	1.48	0.13*	275	88
Norway spruce <i>n</i> =5	1.34*	0.74	1.30	0.12*	1183*	87

*Significantly different from hardwoods at 5% level.

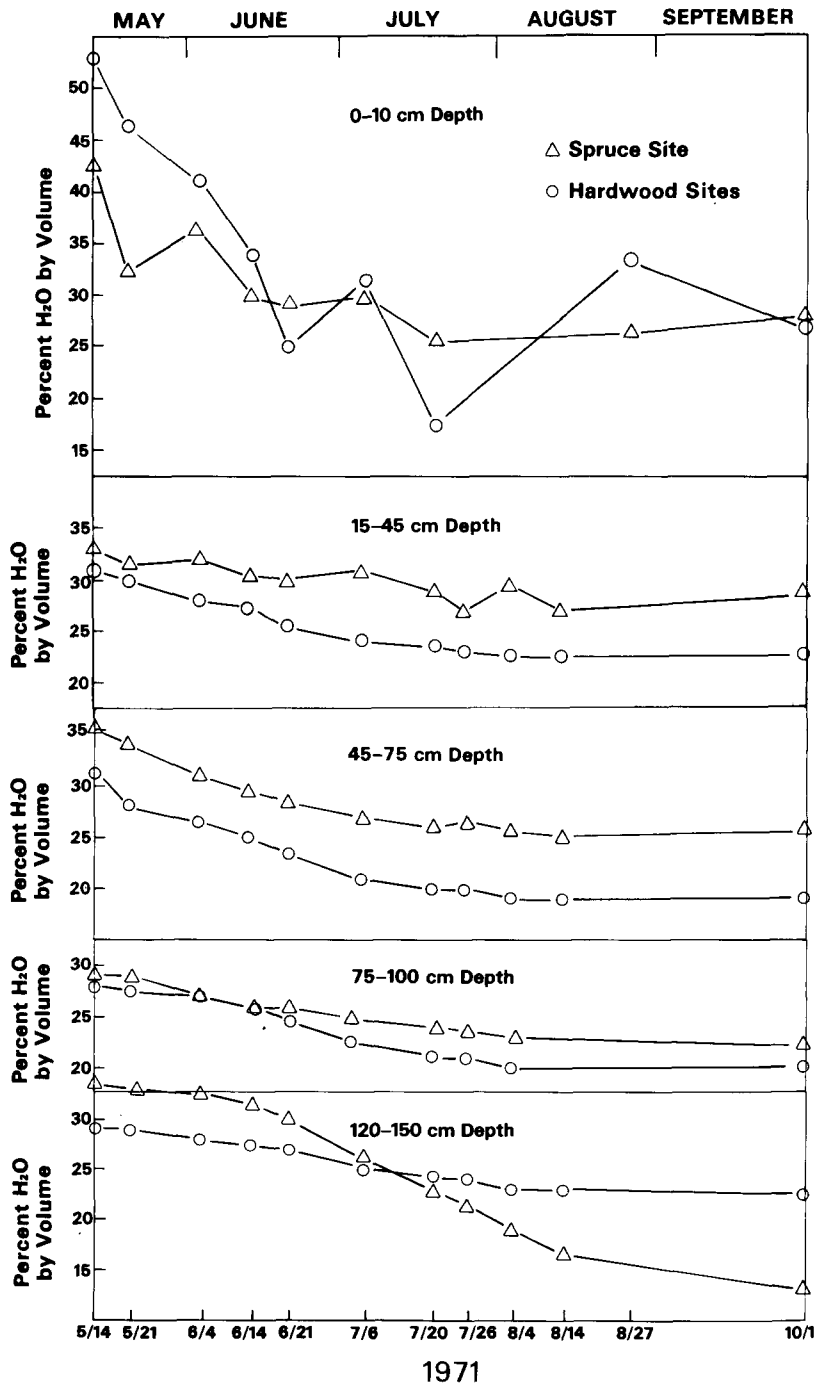


Fig. 1. Comparison of spruce and hardwood soil moisture regimes at several depths.

TABLE III

Comparative soil properties in the upper 15 cm of soil

Soil depth (cm)	pH	Extractable Al ($\mu\text{g/g}$ soil)		Extractable Mn ($\mu\text{g/g}$ soil)		Extractable $\text{NH}_4\text{-N}$ ($\mu\text{g/g}$ soil)		$\text{NO}_3\text{-N}$ ($\mu\text{g/g}$ soil)	
		Spruce	Hardwoods	Spruce	Hardwoods	Spruce	Hardwoods	Spruce	Hardwoods
0-15	5.0*	6.2	136.0	80.9	24.7*	15.1	11.1	7.4	2.2
25	5.6	5.4	135.0	160.8	3.1	1.8	3.3	4.5	1.5
									0.8
Soil depth (cm)	pH	Exchangeable Ca meq/100 g		Exchangeable Mn meq/100 g		Exchangeable K meq/100 g		Exchangeable H meq/100 g	
		Spruce	Hardwoods	Spruce	Hardwoods	Spruce	Hardwoods	Spruce	Hardwoods
0-15	5.0	6.2	6.5	8.7	2.8	1.6	0.3	0.3	9.7
25	5.6	5.4	9.4	4.6	4.5	2.5	0.3	0.3	3.7
									12.1

*Significantly different from hardwoods at 5% level.

Note: nitrogen averages based on eight bi-weekly samplings between 15 June and 14 October 1972; other averages based on single-date pit samples from three spruce site pits and nine hardwood site pits.

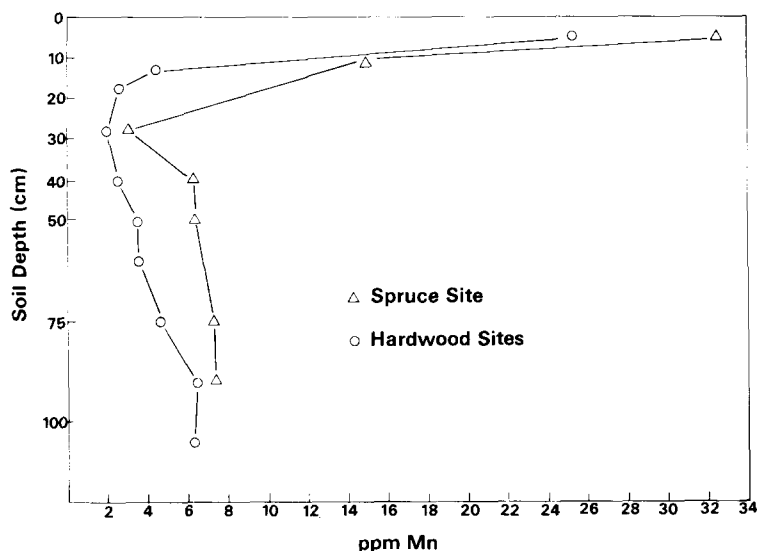


Fig. 2. Comparison of soil profile distributions of extractable manganese between spruce and hardwood ecosystems.

DISCUSSION

Graphed soil moisture percentages at most depths for the three hardwood sites were tightly clustered and consistently separated from those at the spruce site during the period of measurement in 1971. Since soil texture differences (Fig. 3A) do not account for these divergent soil moisture regimes, the regime differences are considered to be a function of the vegetation differences. Because of soil similarities and the fact that regime data are in accord with comparative soil moisture data derived from investigations by other workers, some interpretations seem warranted.

Higher surface soil moisture percentages at the hardwood sites prior to mid-June must certainly have been partly a result of limited transpiration by the deciduous hardwood trees and their deciduous understories prior to, and during, their leafing out from early May to early June.

Greater fluctuations in the surface soil moisture at the hardwood sites after mid-June suggest the effect of greater surface soil root concentrations at those sites and/or less ability of the hardwoods to draw upon deep moisture supplies during periods of high soil moisture tensions in the surface soil. Sugar maple, white oak, and northern red oak all exhibit a dense network of fine roots in the surface horizons of these soils. Further, the roots of several hardwood understory species are confined to this zone. Spruce roots of all sizes are also concentrated there, but no understory plants are present. Since no quantitative comparisons have been made between the surficial spruce roots and those roots at the hardwood sites, no causal conclusions can be drawn.

Higher soil moisture percentages within the 15–75 cm depth zone at the

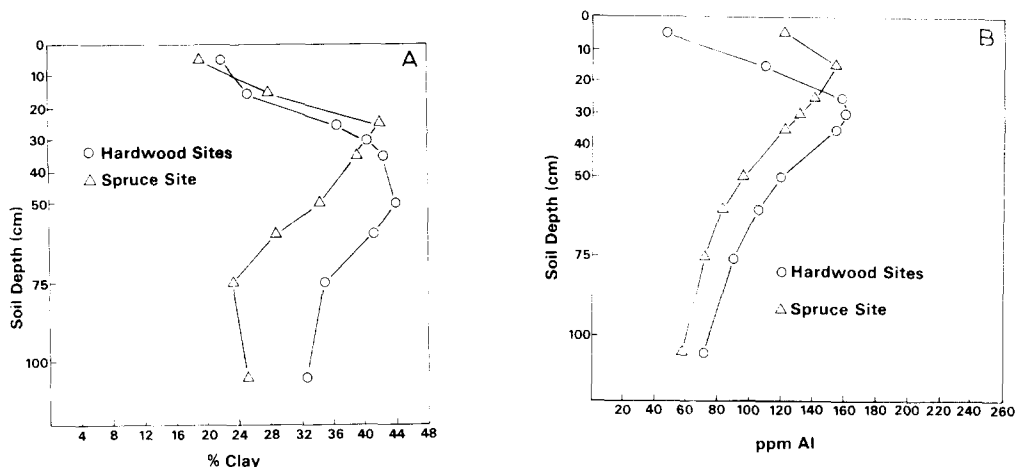


Fig. 3. A, Comparison of soil profile distributions of % clay between spruce and hardwood ecosystems. B, Comparison of soil profile distributions of extractable aluminum between spruce and hardwood ecosystems.

spruce site points to a relative paucity of spruce roots in this zone. In corroboration of this view, soil pit observations recorded in a previous section of this report reveal no roots larger than 15 mm in this zone, and those smaller than 15 mm were only sparsely distributed below a depth of 35 cm. In contrast, large, horizontally spreading secondary roots of the oaks are clearly concentrated in the 15–75 cm depth zone, a distribution repeatedly observed in northeastern Illinois soils. This zone is leached and contains the highest illuvial clay and extractable aluminum concentrations in the soils at the spruce and hardwood sites (Fig. 3). No other high or low concentrations of measured soil properties correspond closely with this low soil moisture depletion zone at the spruce site. At the lower boundary of this zone, the soil is unleached (average depth to effervescence due to carbonates is exactly 75 cm) and clay percentages are similar to those in the upper 15 cm. Wilde (1958) reports that high concentrations of soluble aluminum in illuvial horizons of podzolized soils appear to arrest the downward penetration of tree roots, and research with other plants has shown that aluminum can inhibit lateral root elongation (Rios and Pearson, 1964). Bimorphic root systems are characteristic of strongly leached soils with surface horizons enriched in organic matter (Wilde, 1958). In the bimorphic root systems of Norway spruce depicted by Köstler (Mitscherlich, 1970), the vertical joiners or sinker roots extend through the 30–75 cm zone in a relatively unbranched condition.

Lack of critical values of extractable aluminum for spruce root branching or any data on fine root accumulation of aluminum precludes plausible emphasis on this possible relationship. Because the spruce roots below a depth of 15 cm tend to be confined to structural ped surfaces and glacial till fractures similar to the cases reported by Taylor (1974), mechanical resistance may be more reasonably suspected as a limiting factor, especially considering

the high clay content (28–42%) and an average bulk density of 1.5 g/cc within the 15–75 cm zone.

The intensity and magnitude of soil moisture depletion below a depth of 120 cm at the spruce site is taken as clear evidence of appreciable spruce root concentration there. In the soil pits, vertical stringers of small (diameter < 5 mm) roots along glacial till fracture planes were present to a depth of at least 150 cm. The soil material below a depth of 75 cm is designated as C horizon or relatively unaltered glacial till, and is dense (bulk density of 1.8) and calcareous (CaCO_3 equivalent > 30%). These conditions are inimical to the root growth of many plants because of the associated poor aeration and low solubilities of certain nutrients. That they are not impenetrable barriers to Norway spruce roots is supported by European forestry experience (Wilde, 1958).

Rapid loss of moisture from the 120–150 cm began during the week of 14–21 June 1971, a week characterized by abnormally high temperatures and only 7.1 mm of rain at the Arboretum. During this week, depletion of soil moisture at the hardwood sites increased at rates which gradually decreased with soil depth. At the spruce site, the only temporal increase in depletion rate occurred in the 120–150 cm soil depth zone, suggesting that fine root extension or water absorption capacity of the surface roots may have been impaired by a decline in soil moisture from 36% to 30% (by volume) during the previous 10 days as compared to their deeper roots and the surface roots at the hardwood sites. Regardless of the exact reason for this downward shift of activity within the spruce root system, the rapid depletion rate at the lower level remained constant until 14 August despite minor recharge episodes in overlying horizons.

Though extractable Al concentrations in the upper 15 cm of soil were higher at the spruce site, they were not significantly different from those of the hardwood sites. The effect of red and white pine plantations on extractable aluminum in the same soil type seems to be much more pronounced in a much shorter time (Messenger, 1978). Those differences can be largely attributed to the high aluminum biocycling capacity of the pines (380 ppm in old needles) compared to that of the spruce (88 ppm in old needles).

The apparent effects of the spruce plantation on the low pH in the upper 15 cm of soil may be explained in part by a relatively low rate of magnesium cycling during the nearly 50 years of spruce occupancy as suggested by the foliar data in Table II, but not corroborated by the percent exchangeable Mg calculated from Table III. The greatest contribution to the low pH, however, may be from organic acids, since spruce litter is known to produce extracts with high concentrations of these acids (Stevenson, 1967). From whatever cause, a soil pH below 5.5 is favorable to manganese solubility, thus offering a partial explanation for the comparatively high values of extractable Mn in the surface soil at the spruce site. Deemed equal in importance, however, is the high foliar manganese contributions by Norway spruce. High foliar manganese may be typical for this species since Ovington (1956) also found high values compared to hardwoods growing on the same soil.

A favorable pH for Mn solubility does not explain the relatively high extractable Mn concentrations deeper in the soil at the spruce site, however. Sauchelli (1969) states that organic matter and reducing conditions also favor Mn solubility. The positive effect of organic matter may be functional throughout the soil profile at the spruce site because of the intensive production of organic acids which favor the formation of soluble complexes or organic constituents with metals, including Mn. Once formed, some of these soluble complexes, such as chelates, are quite mobile and even remain soluble where the soil pH is neutral to alkaline, especially if reducing conditions are prevalent (Page, 1962; Drake, 1964; Stevenson, 1967; Norvell, 1972; Stevenson and Ardakani, 1972). Thus the secondary maximum of extractable Mn in the subsoil at the spruce site (Fig. 2) is likely favored by reducing conditions resulting from the fine texture and persistence of high soil moisture percentages previously discussed. In summary, it is believed that Norway spruce litter is a significant source of soil Mn and may contribute to its solubilization, at least in part by the release of chelating agents such as those thought to facilitate iron oxide removal from soils supporting Norway spruce (Epstein, 1972).

Acidic surface soil conditions and a high C/N ratio favor a microfloral dominance by fungi and low nitrification rates (Corke, 1958; Messenger et al., 1972). This type of microfloral composition logically leads to a certain amount of ammonium accumulation such as that characterizing the surface soil in the spruce plantation, thus pointing to ammonium as the major current source of available N. Foliar samples taken from the study sites showed much lower nitrogen concentrations in the needles of both species of spruce than in any of the hardwood leaves. If annual litter fall is comparable, the resultant lower rate of N cycling in the spruce plantation would be expected to maintain a higher C/N ratio in the upper humic horizons of the soil.

While most of the chemical soil differences between the hardwood and spruce ecosystems can be logically explained by differential effects imposed by different species compositions, an additional factor that may have contributed to the surface soil differences shown in Table II is erosion under row crop farming prior to plantation establishment. Since there is testimony indicating little evidence of erosion at the time of plantation establishment, and since the depths to the bottom of the B2 horizons at the spruce sites were all within the ranges exhibited at the hardwood sites, the erosion factor is assumed to have been minimal.

CONCLUSIONS

The 1971 soil moisture regime in the spruce plantation differed from regimes in adjacent native hardwood forests by exhibiting a drier surface soil in May, a moister B horizon from late spring to fall, and a much greater C horizon depletion rate to a depth of 1.5 m from mid-June to mid-August resulting in much drier conditions in that horizon from early August through

October 1. These distinctive trends are consistent with on-site rooting pattern observations and results of certain previous investigations.

The significantly greater soil acidity and extractable soil Mn in the spruce plantation can be partly explained by the low foliar Mg and high foliar Mn of the spruces as compared to the hardwoods. However, extrapolations of previous research dictate the need to study the soil acidity implications of the low-nitrogen effects of spruce litter on microbial populations and decomposition products.

Extractable surface soil Al is not as high in the spruce plantation or in the native oak forests as is in red pine and white pine plantations on the same soil type. Neither the spruces nor the hardwoods have the aluminum biocycling power exhibited by the pines.

SILVICULTURAL AND ARBORICULTURAL IMPLICATIONS

There is a fund of practical experience concerning the suitability of various tree species for heavy, difficultly penetrable soils. On such soils, a high root activity and a low demand for exogenous oxygen are decisive factors in determining site tolerance of a tree species (Lyr and Hoffmann, 1967). These attributes apparently apply to Norway spruce, and it is therefore assumed that the indications of deep rooting at the mixed spruce site primarily reflect the activity of this species. Deeper rooting of Norway spruce compared to Colorado blue spruce may partially explain the less rapid growth rate of the latter species. Based on the above considerations, planting of Norway spruce on land dominated by Morley silt loam or similar soils is considered to be more silviculturally and arboriculturally sound than planting certain other tree species such as red pine (*Pinus resinosa*), which roots poorly and soon stagnates on such soils (Van Eck and Whiteside, 1963).

In addition to the favorable growth rate of Norway spruce, the probability that the roots of this species penetrate into the dense, calcareous C horizon is of long-range silvicultural importance since the resultant root channels left by one rotation may permit the roots of a replacement species to penetrate an otherwise inhospitable zone.

Where Norway spruce exhibits its bimorphic root system and does not utilize much of the available soil moisture throughout an appreciable depth of soils as indicated in the present study, more intensive site utilization with minimal root competition between species may be possible by interplanting or underplanting species which normally develop extensive root systems at soil depths sparsely occupied by spruce roots. Underplantings of *Taxus cuspidata* have been successfully used at the Morton Arboretum.

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